Writing Growth and Decay Models from Data

Answer Key

Algebra 1

Quick Answers

1. (a) the multiplier = 1.5, (b) prediction at t = 5 = 455.63
2. (a) the multiplier = 0.8, (b) percent lost each step = 20
3. (a) the multiplier = 1.08, (b) count after 6 hours = 396.72
4. (a) the multiplier = 0.85, (b) value after 4 years = 9396.11
5. (a) the hourly multiplier = 0.5, (b) amount after 5 hours = 25
6. years to reach 4800 = 20
7. (a) the daily multiplier = 3, (b) count on day 4 = 3240
8. hours until 20 mg remain = 5 hr
9. (a) the multiplier = 0.97, (b) population after 10 years = 8849.09
10. (a) mass after 24 days = 150 g, (b) days until 37.5 g remain = 40, (c) explanation = —

What is verified

18 of 19 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSF-LE.A.1 – *problems 1, 2, 3, 4, 5, 7, 9*
HSF-LE.A.4 – *problems 6, 8, 10*
Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

4. *If they got 7200: subtracted 15 percent of the original price four times instead of compounding*

1. Colony table $60, 90, 135, 202.5$. (a) The hourly multiplier. (b) Predict the count at $t = 5$.

Solution (a): Divide consecutive counts: $90 \div 60$. The same ratio appears again ($135 \div 90$), so the model is exponential. $\boxed{r = 1.5}$

Solution (b): The model is $60 \cdot 1.5^t$. At $t = 5$: $1.5^5 = 7.59375$, so 60×7.59375. $\boxed{455.63}$

2. Yearly table $800, 640, 512, 409.6$. (a) The multiplier. (b) Percent lost each year.
Solution (a): $640 \div 800$. $\boxed{r = 0.8}$
Solution (b): A multiplier of 0.8 keeps 80% and so loses the rest: $100(1-0.8) = 100(0.2)$. $\boxed{20}$

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3. Culture of 250 growing 8% an hour. (a) The multiplier. (b) Count after 6 hours.
Solution (a): Growth of 8% keeps the whole and adds 8 hundredths: $1 + \frac{8}{100}$. $\boxed{r = 1.08}$

Solution (b): $1.08^6 = 1.58687432\ldots$, so $250 \times 1.58687432 = 396.71858\ldots$, rounded to two decimal places. $\boxed{396.72}$

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4. Car worth \$18,000 losing 15% a year. (a) The multiplier. (b) Value after 4 years.
Solution (a): Losing 15% leaves 85%: $1 - \frac{15}{100}$. $\boxed{r = 0.85}$
Solution (b): $0.85^4 = 0.52200625$, so 18000×0.52200625. $\boxed{9396.11}$
Common wrong answer: \$7200 comes from subtracting 15% of the *original* price four times. Each year's loss is taken from the value at the start of that year, which is smaller every time.

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5. Sample: 800 mg at $t = 0$, 100 mg at $t = 3$ hours. (a) The hourly multiplier. (b) Mass after 5 hours.
Solution (a): Three hours of multiplying give $800r^3 = 100$, so $r^3 = \frac{100}{800} = \frac{1}{8}$. The number whose cube is $\frac{1}{8}$ is $\frac{1}{2}$. $\boxed{r = 0.5}$
Solution (b): $800 \cdot \left(\frac{1}{2}\right)^5 = \frac{800}{32}$ — halve five times: $400, 200, 100, 50$. $\boxed{25}$

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6. Population $300 \cdot 2^{t/5}$. When does it reach 4800?
Solution: Divide by 300: $2^{t/5} = 16 = 2^4$, so $\frac{t}{5} = 4$. Check: $300 \cdot 2^4 = 300 \cdot 16 = 4800$ ✓ $\boxed{t = 20}$

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7. Bacteria: 40 on day 0, 120 on day 1. (a) The daily multiplier. (b) Count on day 4.
Solution (a): One day of multiplying takes 40 to 120, so $r = 120 \div 40$. $\boxed{r = 3}$
Solution (b): The model is $40 \cdot 3^t$, and $3^4 = 81$, so 40×81. $\boxed{3240}$

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8. Drug: $640 \cdot \left(\frac{1}{2}\right)^t$ mg. When is 20 mg left?
Solution: Divide by 640: $\left(\frac{1}{2}\right)^t = \frac{20}{640} = \frac{1}{32}$, and $\frac{1}{32} = \left(\frac{1}{2}\right)^5$. Check: halving 640 five times gives $320, 160, 80, 40, 20$ ✓ $\boxed{t = 5 \text{ hr}}$

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9. Town of 12,000 losing 3% a year. (a) The multiplier. (b) Population after 10 years.
Solution (a): $1 - \frac{3}{100}$. $\boxed{r = 0.97}$

Solution (b): $0.97^{10} = 0.73742412\ldots$, so $12000 \times 0.73742412 = 8849.0895\ldots$, rounded to two decimal places. $\boxed{8849.09}$

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10. Radioactive sample: $1200 \cdot \left(\frac{1}{2}\right)^{t/8}$ grams, half-life 8 days. (a) Mass at 24 days. (b) When is 37.5 g left? (c) Why halving is not subtracting.

Solution (a): 24 days is three half-lives, so $\frac{t}{8} = 3$ and the mass halves three times: $1200 \to 600 \to 300 \to 150$. $\boxed{150\text{ g}}$

Solution (b): Divide by 1200: $\left(\frac{1}{2}\right)^{t/8} = \frac{37.5}{1200} = \frac{1}{32} = \left(\frac{1}{2}\right)^{5}$, so $\frac{t}{8} = 5$. Check: five half-lives from 1200 give $600, 300, 150, 75, 37.5$ ✓ $\boxed{t = 40}$

(c) — instructor-judged. A correct answer says halving takes the same *fraction* of whatever is present, so each successive loss is smaller in absolute terms and the mass approaches zero without reaching it; subtracting a fixed number of grams each period would remove the same amount every time and would drive the mass to zero and then below it, which is impossible. Full credit for contrasting a constant fraction with a constant amount and drawing the consequence; partial credit for naming the difference without the consequence.